

Birla Institute of Technology & Science, Pilani
K.K. Birla, Goa Campus

Theory of Computation (CS F351) Semester – I, Year: 2025-2026
Mid-Semester Test

Date: 09-10-2025, Duration: 90 mins., Max. Marks: 60, Weightage: 30% (Closed Book)

Q1. Which of the following statements are true/false? Give reasons/counterexamples

- a) If L_1 and L_2 are nonregular, then $L_1 \cup L_2$ is nonregular.
- b) $L = \{w \in \{0, 1\}^* : |w| > 3\}$ is a regular language
- c) Language $L = \{a^i b^j : \gcd(i, j) = 1\}$ is regular.
- d) If L is a regular language, then so is $\{w : w \in L \text{ and } w^R \in L\}$, where w^R denotes the reversal of string w .

[20 Marks]

Q2. Construct a DFA for the language $\{w \in \{a, b\}^* : w \text{ has both } ab \text{ and } ba \text{ as substrings}\}$.

[8 Marks]

Q3. Use Pumping Lemma to show that $L = \{0^i 1^j 0^k : k \geq i + j\}$ is nonregular.

[10 Marks]

Q4.

a) Write regular expression for the language: All strings in $\{a, b\}^*$ with exactly one occurrence of the substring aaa .

[4 Marks]

b) If $L = \{w \in \{a, b\}^* : w \text{ ends in } a\}$. For the relation $\approx_L$, determine and compute the distinct equivalence classes.

[8 Marks]

Q5.

a) Does $S \rightarrow S01S, S \rightarrow S10S, S \rightarrow \epsilon$ generate the language

$L = \{w \in \{0, 1\}^* : no(w) = n_1(w)\}$, where $no(w)$ denotes number of 0's in w .

Justify your answer.

[4 Marks]

b) Let $L = \{w \in \{0, 1\}^* : no(w) = n_1(w) + 1\}$.

Find a context free grammar G such that $L = L(G)$.

[6 Marks]